Points 6 Published

Details Questions

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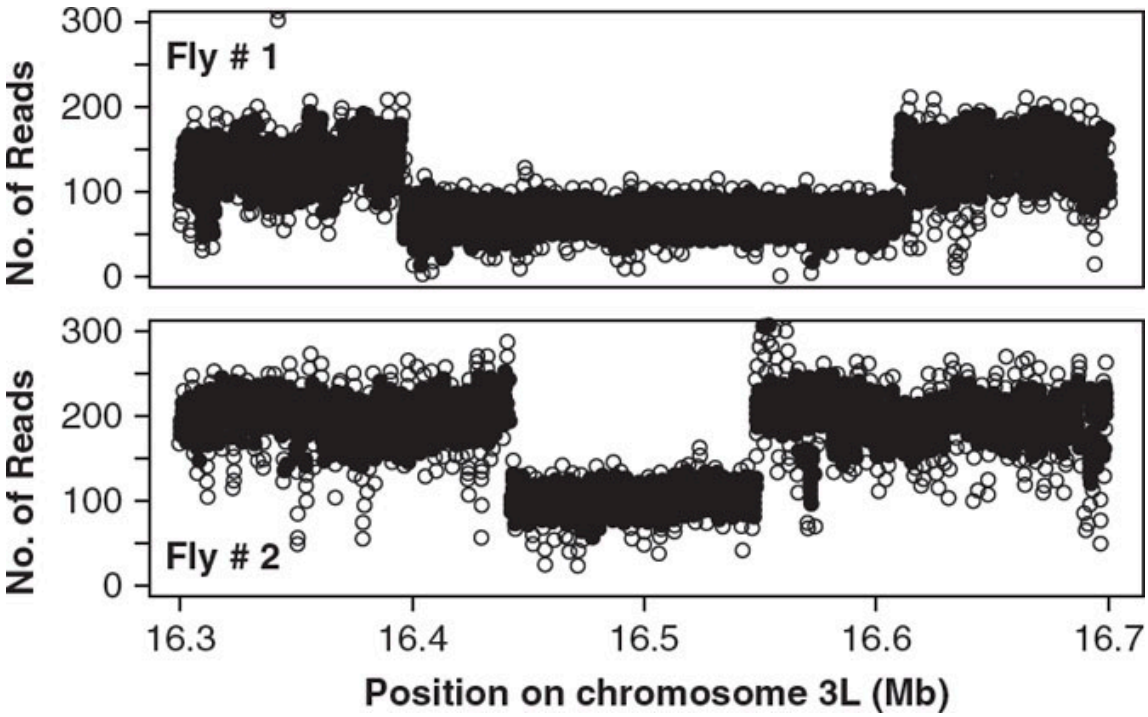
Question

The two graphs below represent genomic sequencing data for two fruit flies with scarlet-colored eyes. Each fly is a heterozygote for a chromosome 3 with normal gene order and a recessive *scarlet* (*st*) mutation, and a rearranged chromosome 3; the rearrangements are different in the two flies. Each circle on the graph indicates the number of sequence reads (*y* coordinate) obtained for a 100bp sequence starting at the position on chromosome 3 that is shown by the *x* coordinate.



Question 0.6 pts

What type of chromosomal rearrangement is occurring in Fly #1?

- Deletion
- Duplication
- Inversion
- Translocation

Question 0.6 pts

What type of chromosomal rearrangement is occurring in Fly #2?

- Deletion
- Duplication
- Inversion
- Translocation

Question 0.6 pts

Can you narrow down the region where the *st* gene is most likely to occur? What are the coordinates?

- 16.3Mb-16.7Mb
- 16.4Mb-16.45Mb
- 16.45Mb-16.55Mb
- 16.55Mb-16.6Mb
- 16.6Mb-16.7Mb

Question

Two wild-type fragments of human genomic DNA from the long arms of two nonhomologous acrocentric chromosomes are shown below. On both fragments the centromere-to-telomere direction is left-to-right. The red lines numbered 1,2, and 3 indicate the positions of breakpoints that could be caused by ionizing radiation. Certain combinations of these breaks could produce chromosomal rearrangements when the broken pieces are stitched back together by DNA repair systems.

```
5'  TCGATTCCGGAAAGCTTAGTTTCCCGGGACGTATTGCCAACCTAGGTAAGCGCCGAATATCCATGGGCACC  3'
3'  AGCTAAGGCCTTTCGAATCAAAGGGCCCTGCATAACGGTTGGATCCATTCGCGGCTTATAGGTACCCGTGG  5'
                   1                                        2

                        5'  GGCAATAGCCTAGGAACTTTTAGGCCAATTAA  3'
                        3'  CCGTTATCGGATCCTTGAAAATCCGGTTAATT  5'
                                           3
```

Question 0.6 pts

Which two of the breakpoints could yield a deletion?

- 1 and 2
- 1 and 3
- 2 and 3

Question 0.6 pts

You will want to diagnose the presence of the rearrangements by using PCR. In every case, one of the primers you will use for the PCR is primer A, whose sequence is:

5' TCGATTCCGGAAAGCT 3'

Which primer sequence, in conjunction with primer A, will allow you to diagnose the presence of a deletion in the patient's genomic DNA. The evidence should be positive, not negative. (i.e. you should see a PCR product. The absence of a band is not informative)

Note the polarity.

- 5' TCGATTCCGGAAAGCT 3'
- 3' AGCTAAGGCCTTTCGA 5'
- 3' TGGATCCATTCGCGGC 5'
- 5' AATATCCATGGGCACC 3'
- 3' TTATAGGTACCCGTGG 5'

Question 0.6 pts

Which two breakpoints could yield an inversion?

- 1 and 2
- 1 and 3
- 2 and 3

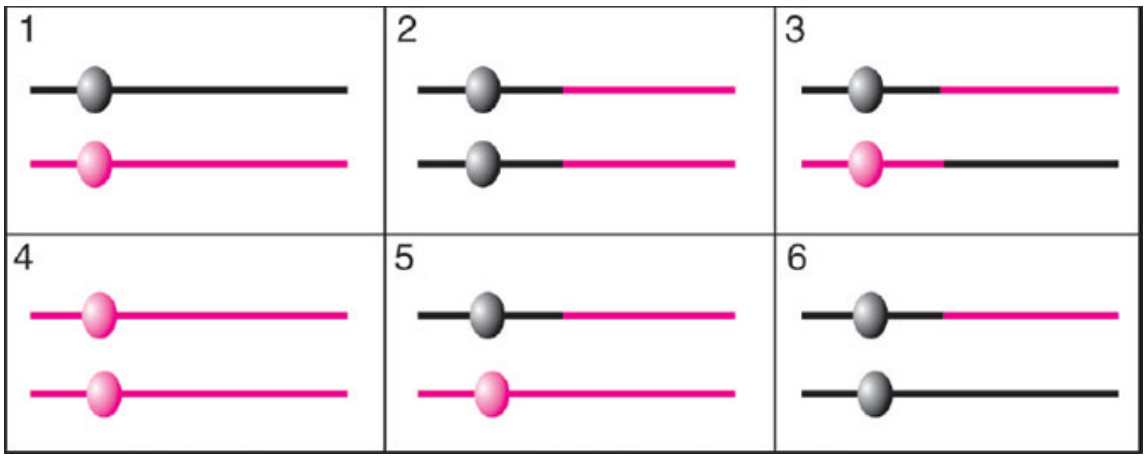
Question 0.6 pts

An inversion occurred between the two breakpoints indicated above. You perform high-throughput sequencing of this genomic DNA. Which of the following reads would support the presence of the inversion event. The breakpoint in the read is indicated with a "|".

- 5' TCGATTCCGGAAAGCT|AATATCCATGGGCACC 3'
- 5' TCGATTCCGGAAAGCT|CTTTTAGGCCAATTAA 3'
- 5' TCGATTCCGGAAAGCT|CGGCGCTTACCTAGGT 3'

Question

In the following figure, black and pink lines represent nonhomologous chromosomes.



For the following descriptions, more than one diagram may correspond to each statement, and a diagram may be a correct answer for more than one question. Assume no crossover events.

Question 0.6 pts

Gametes produced by a translocation heterozygote.

- [ ] 1
- [ ] 2
- [ ] 3
- [ ] 4
- [ ] 5
- [ ] 6

Question 0.6 pts

Genetically balanced gametes produced by a translocation heterozygote.

- [ ] 1
- [ ] 2
- [ ] 3
- [ ] 4
- [ ] 5
- [ ] 6

Question 0.6 pts

Genetically imbalanced gametes that arise from adjacent-2 segregation.

- [ ] 1
- [ ] 2
- [ ] 3
- [ ] 4
- [ ] 5
- [ ] 6

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